

L05: Fundamentals of Machine Learning

CSci 560 Deep Learning w/ Python (Chollet) Ch. 5

Derek Harter

Department of Computer Science
East Texas A&M University

Summer 2025

L05.1 Generalization: The goal of machine learning

CSci 560: Deep Learning w/ Python (Chollet) Ch. 5.1

Derek Harter

Department of Computer Science
East Texas A&M University

Summer 2025

Generalization: The goal of machine learning

Fundamental Issue of ML

The fundamental issue in machine learning is the tension between optimization and generalization.

- After just a few epochs of training, performance on never-before-seen data starts to diverge
 - The models start to **overfit**
- **Optimization** refers to the process of adjusting a model to get the best performance possible on the training data.
- **Generalization** refers to how well the trained model performs on data it has never seen before.

Underfitting and Overfitting

- **underfit** the network hasn't yet modeled all relevant patterns in the training data.
- Eventually validation loss begins to increase while training loss continues to improve
 - Generalization has stopped improving, validation metrics stall and then degrade
- The model is **overfitting**
 - It is learning patterns specific to the training data (noise)

Overfitting is particularly likely to occur when your data is noisy, if it involves uncertainty, or if it includes rare features.

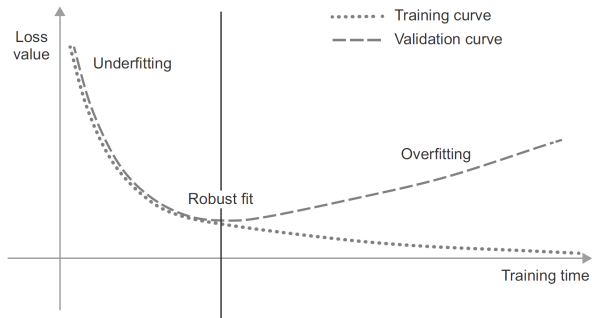


Figure 1: Canonical overfitting behavior. (Chollet, 2021, pg.122)

Noisy Training Data

- In real-world datasets, inputs are often **ambiguous** or **invalid**.
- Even worse, some of the data will be **mislabeled**.
- If a model goes out of its way to incorporate such outliers, its generalization performance will degrade.

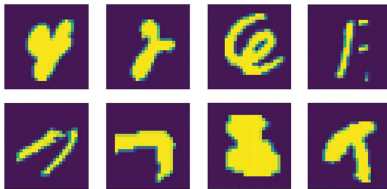


Figure 2: Weird MNIST training samples, what are these? (Chollet, 2021, pg.123)

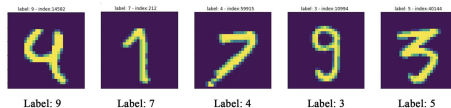


Figure 3: Mislabeled MNIST training samples? (Chollet, 2021, pg.123)

Dealing with Outliers: Robust fit vs. Overfitting

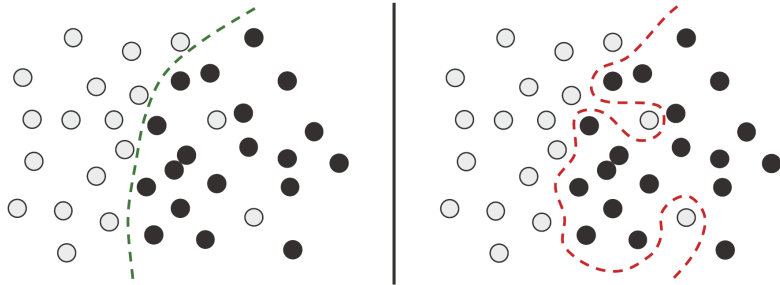


Figure 4: Dealing with outliers: robust fit vs. overfitting. (Chollet, 2021, pg.124)

- If a model goes out of its way to incorporate such outliers (ambiguous, mislabeled, noisy), its generalization performance will degrade.
- On the right, a model that overfits goes out of its way to incorporate examples that may be mislabeled.
- On the left, this model probably will generalize much better by ignoring outliers.



Ambiguous Features

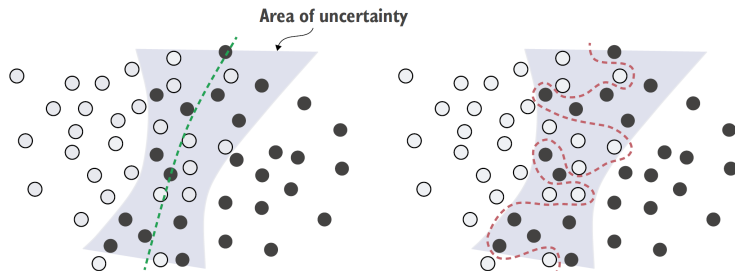


Figure 5: Robust fit vs. overfitting giving an ambiguous area of the feature space. (Chollet, 2021, pg.124)

- Not all data noise comes from inaccuracies
 - Even clean and correctly labeled data can be noisy.
- For example, the area of uncertainty might represent noise where it is difficult to differentiate between the two classes.
- A model that overfits on such noisy data will be too confident about ambiguous regions of the feature space.

Rare Features and Spurious Correlations

- Machine learning models trained on datasets that include rare features are highly susceptible to overfitting.
- Spurious correlations are present even in noise.
- Example training MNIST and adding noise vs. just 0's.

The Nature of Generalization in Deep Learning

- A deep learning model can be trained to fit anything, as long as it has enough representational power.
- You can get training loss on completely random noise. (see example)
- How can deep learning models generalize?
 - When learning noise like this example, are basically learning an ad hoc mapping between training inputs and targets, like a fancy dict

The Manifold Hypothesis

Manifold Hypothesis

The **Manifold**

Hypothesis posits that all natural data lies on a low-dimensional manifold within the high-dimensional space where it is encoded.

- Take MNIST dataset, total possible inputs is $256^{28 \times 28}$
- Very few of these inputs would look like a digit.
 - actual digits occupy a tiny *subspace* of the possible space
 - not a random subspace, it is **continuous**
 - all samples in the subspace are *connected* by smooth paths
- Handwritten digits form a **manifold** within the space of possible inputs.

The manifold hypothesis implies that:

- Machine learning models only have to fit relatively simple, low-dimensional highly structured subspaces within their potential input space.
- It's always possible to **interpolate** between two inputs.

The ability to interpolate between samples is the key to understanding generalization in deep learning.

Interpolation as a Source of Generalization

- You can make sense of the **totality** of the space using only a **sample** of the space.
- Interpolation along a manifold is different from linear interpolation in the parent space.
- Interpolation can only help you make sense of things that are close to what you've seen before.



Manifold interpolation
(intermediate point
on the latent manifold)



Linear interpolation
(average in the encoding space)

Figure 6: Difference between linear interpolation and interpolation on the latent manifold. Every point on the latent manifold of digits is a valid digit, but the average of two digits usually isn't. (Chollet, 2021, pg.124)

Why Deep Learning Works



Figure 7: Uncrumpling a complicated manifold of data. (Chollet, 2021, pg.130)

- A deep learning model is a tool for uncrumpling paper balls, that is for disentangling latent manifolds.
- A deep learning model is basically a very high-dimensional curve
 - smooth and continuous (since differentiable)
 - curve fitted to data via gradient descent (smoothly and incrementally)

Why Deep Learning Works

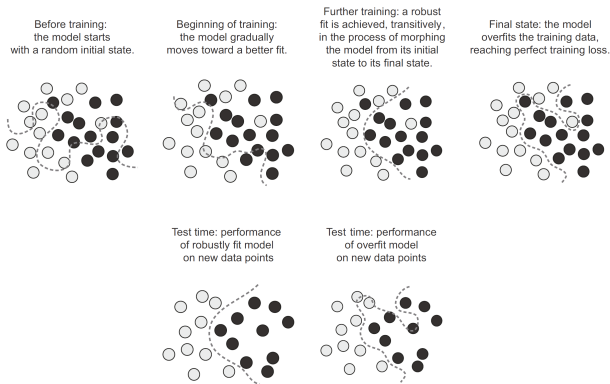
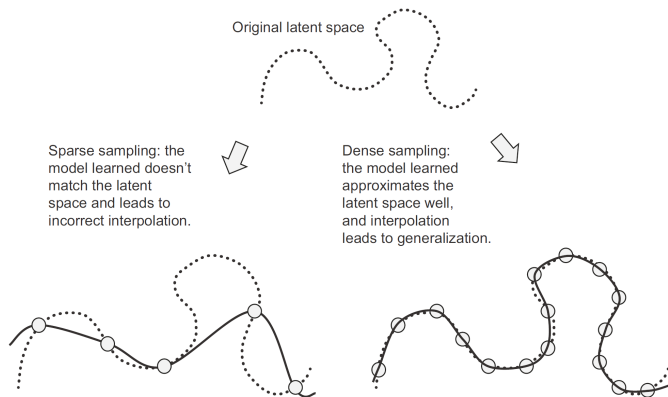


Figure 8: Going from a random model to an overfit model, and achieving a robust fit as an intermediate state. (Chollet, 2021, pg.131)

- The curve involves enough parameters that it could fit anything (even noise)
- **At some intermediate point during training the model roughly approximates the natural manifold of the data.**
- Deep learning models implement a smooth, continuous mapping from inputs to outputs. (differentiable)
- Deep learning models structured in a way that mirrors *shape* of the information in training data.



Training Data is Paramount



- The power to generalize is a consequence of the natural structure of your data.
- Data curation and feature engineering are essential.
- For a model to perform well **it needs to be trained on a dense sampling of its input space.**
- The best way to improve a deep learning model is to train it on more data or better data.

Figure 9: A dense sampling of the input space is necessary in order to learn a model capable of accurate generalization. (Chollet, 2021, pg.131)

L05.2 Evaluating Machine Learning Models

CSci 560: Deep Learning w/ Python (Chollet) Ch. 5.2

Derek Harter

Department of Computer Science
East Texas A&M University

Summer 2025

L05.3 Improving Model Fit

CSci 560: Deep Learning w/ Python (Chollet) Ch. 5.3

Derek Harter

Department of Computer Science
East Texas A&M University

Summer 2025

L05.4 Improving Generalization

CSci 560: Deep Learning w/ Python (Chollet) Ch. 5.4

Derek Harter

Department of Computer Science
East Texas A&M University

Summer 2025

Chollet, F. (2021). *Deep learning with python* (second). Manning.